# Week 3

## Lecture 6 Properties of finite-dimensional vector spaces

*This covers part of **Subsections 1.3.2** and all of **1.3.3** in the course notes.*

Fix a vector space $V$ over $\mathbb{F}$. We saw that $V$ is finite-dimensional when it has a finite basis, in which case any two bases have the same size. We called this common size the *dimension* of $V$ and denoted it by $\dim(V)$.

Let us assume throughout the lecture that $\dim(V) = n$. Then we can describe the spanning and/or independence behavior of any finite set of vectors depending on its size relative to $n$. We state this as a theorem to mark its importance.

**Theorem 6.1**

Let $S \subseteq V$ be a finite subset. Denote by $|S|$ the size of the set $S$ (i.e. the number of elements belonging to $S$).

**(i)** If $|S| > n$, then $S$ must be linearly dependent.

**(ii)** If $|S| < n$, then $S$ cannot be a spanning set for $V$.

**(iii)** If $|S| = n$, then $S$ is a spanning set for $V$ if and only if $S$ linearly independent.

*Proof.* **(i)** is precisely the contrapositive of **Lemma 5.2** (where in the statement of this lemma, we choose our independent set $T$ to be any basis for $V$).

For **(ii)**, suppose $S$ is a spanning set for $V$. Then by **Lemma 5.1** there exists a linearly-independent subset $T$ of $S$ with $\text{Span}(T) = \text{Span}(S) = V$. But this means $T$ is a basis for $V$, and so $T$ has $n$ vectors, thus $S$ has at least $n$ vectors.

For **(iii)**, if $S$ is a spanning set of $n$ vectors but it is not independent, then there is some $v \in S$ which is a linear combination of the other vectors in $S$. This means $\text{Span}(S \setminus \{v\}) = \text{Span}(S) = V$. So $S \setminus \{v\}$ is a spanning set with $< n$ elements, which contradicts **(ii)**.

Conversely if $S$ is a linearly-independent set of $n$ vectors but it is not a spanning set, then there is some $\vec{v} \in V$ with $\vec{v} \notin \text{Span}(S)$. This means $S \cup \{\vec{v}\}$ is a linearly-independent set with $> n$ elements, which contradicts **(i)**. □

**Remark 6.2**

A nice way to look at what this theorem is telling us is the following: if $V$ is finite-dimensional, then

- a basis for $V$ is a minimal spanning set for $V$,
- a basis for $V$ is a maximal linearly-independent subset of $V$,
- the dimension of $V$ is the smallest number of vectors needed to span $V$, and
- the dimension of $V$ is the largest number of vectors which can be linearly

independent.

The proof of **Theorem 6.1 (iii)** illustrated two general procedures we can carry out:

- removing redundant vectors of a spanning set $S$ to obtain a linearly-independent set $T$ which is also spanning; and
- adding more vectors to a linearly-independent set $T$ until we obtain a spanning set $S$ which is also linearly independent.

In each procedure above, we obtain a basis for $V$! So we have two ways in which we can find a basis for a given vector space. We put these together in the following theorem:

**Theorem 6.3 (Finding bases from spanning/independent sets)**

Given that $V$ is finite-dimensional:

- any spanning set $S$ for $V$ contains a basis $\mathcal{B} \subseteq S$ for $V$; and
- any linearly-independent subset $S$ of $V$ can be extended to a basis $\mathcal{B} \supseteq S$ for $V$.

Let's view some examples to see these processes in action.

**Example 6.4**

Consider the vector space $\mathcal{P}_3(\mathbb{F})$ and the set $S = \{1 + x, 1 - x^2\}$. It is easy to check that this set is linearly independent. We seek to add polynomials to $S$ to obtain a basis. We know $\dim(\mathcal{P}_3(\mathbb{F})) = 4$, so we just need to add two more.

Notice that $1 + x, 1 - x^2 \in \mathcal{P}_2(\mathbb{F})$, which means $\text{Span}(S) \subseteq \mathcal{P}_2(\mathbb{F})$. Hence, $x^3 \notin \text{Span}(S)$. So let $S_1 = \{1 + x, 1 - x^2, x^3\}$ and we can be sure that $S_1$ is still linearly independent.

Now any element in $\text{Span}(S_1)$ is of the form $a(1 + x) + b(1 - x^2) + cx^3$ for $a, b, c \in \mathbb{F}$, and this can be rewritten as:

$$(a + b) + ax - bx^2 + cx^3$$

Is there anything in $\mathcal{P}_3(\mathbb{F})$ which is *not* of the above form? Notice that if $x$ was expressible in the above form, we would have:

$$\begin{aligned} &(a + b) + ax - bx^2 + cx^3 = x \\ \implies\ &(a + b) + (a - 1)x - bx^2 + cx^3 = 0 \end{aligned}$$

We know that the set $\{1, x, x^2, x^3\}$ is independent, so we must have $a + b = a - 1 = -b = c = 0$. But then $b = 0$ and $a = 1$, yet $0 = a + b = 1 + 0 = 1$, which is a contradiction. Therefore, $x$ is not expressible in the above form, so $x \notin \text{Span}(S_1)$. Thus, the set $\mathcal{B} = \{1 + x, 1 - x^2, x^3, x\}$ is independent. Since it has $4 = \dim(V)$ elements, we conclude that $\mathcal{B}$ is a basis for it.

**Example 6.5**

Consider $\mathcal{M}_2(\mathbb{F})$ and the set:

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \right\}$$

We shall accept the fact that $\text{Span}(S) = \mathcal{M}_2(\mathbb{F})$ (and we will see how this can be proved in the next lecture).

Since $\dim(\mathcal{M}_2(\mathbb{F})) = 4$ and $S$ has 5 elements, we need to remove just one of the elements (ensuring that the resulting set will still be a spanning set). We must therefore choose an element of $S$ which is a linear combination of the others. Notice that:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + 2\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} - 2\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix} = 3\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

which means:

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \frac{2}{3}\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} - \frac{2}{3}\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

This proves $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ is in the span of the other elements of $S$, so removing it from $S$ gives:

$$\left\{ \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \right\}$$

which is a basis for $\mathcal{M}_2(\mathbb{F})$.

We have seen that a number of examples of vector spaces appear as subspaces of larger vector spaces. It would be nice to ensure in the abstract that subspaces of a finite-dimensional space will also enjoy finite-dimensionality. The following fact says that it is indeed so, and with more information.

**Proposition 6.6 (Subspace of a finite-dimensional space)**

Let $U$ be a subspace of $V$. Then $U$ is finite-dimensional. Moreover, $\dim(U) \leq n$ and we have $\dim(U) = n$ if and only if $U = V$.

*Proof.* We first show that $U$ is finite-dimensional. If $U = \{\vec{0}\}$, we are done. So assume $U \neq \{\vec{0}\}$. Then there exists some $\vec{u}_1 \in U$ with $\vec{u}_1 \neq \vec{0}$. In particular, $S_1 = \{\vec{u}_1\}$ is linearly independent.

If $U = \text{Span}(\{\vec{u}_1\})$, we are again done. Otherwise, there must be some $\vec{u}_2 \in U$ with $\vec{u}_2 \notin \text{Span}(\{\vec{u}_1\})$. In particular, $S_2 = \{\vec{u}_1, \vec{u}_2\}$ is linearly independent.

Continuing in this way, for each $k \geq 1$ we have $S_k = \{\vec{u}_1, ..., \vec{u}_k\}$ is linearly independent and either $\text{Span}(S_k) = U$ or there is another $\vec{u}_{k+1} \in U$ having $\vec{u}_{k+1} \notin \text{Span}(S_k)$.

But this process must terminate before or at $k = n$ since $\dim(V) = n$. If it didn't, then $S_{n+1} = \{\vec{u}_1, ..., \vec{u}_{n+1}\}$ is a linearly-independent subset of $U$, therefore of $V$, which is impossible. Hence, there must exist some $1 \leq k \leq n$ with $\text{Span}(S_k) = U$, so $U$ is finite-dimensional.

This argument also proves in particular that $\dim(U) = k \leq n$. It is easy to see that $k = n$ if and only if $U = V$. □

**Exercise 6.7**

Compute the dimensions of the subspaces $\mathcal{U}_n(\mathbb{F})$ and $\mathcal{D}_n(\mathbb{F})$ of $\mathcal{M}_n(\mathbb{F})$.

(Hint: consider the standard basis $\{E_{i,j} : 1 \leq i, j \leq n\}$ for $\mathcal{M}_n(\mathbb{F})$. Can you find bases for $\mathcal{U}_n(\mathbb{F})$ and $\mathcal{D}_n(\mathbb{F})$ using elements from here?)

Given a subspace $U$ of $V$ with $\dim(V) = n$, we know that $\dim(U) \leq n$. But it isn't always easy to find a basis for $U$ (unlike the preceding exercise).

**Example 6.8**

Consider the space $\mathcal{M}_2(\mathbb{F})$ of all $2 \times 2$ matrices over $\mathbb{F}$. Let:

$$U = \{A \in \mathcal{M}_2(\mathbb{F}) : \text{Tr}(A) = 0\}$$

where $\text{Tr}(A)$ denotes the *trace* of $A$ (the sum of the diagonal elements of $A$). Notice that the zero matrix $O_{2\times 2} \in U$ since the trace is $0 + 0 = 0$. Moreover, if $A, B \in U$ and $\alpha \in \mathbb{F}$, then by properties of the trace:

$$\text{Tr}(\alpha A + B) = \alpha \,\text{Tr}(A) + \text{Tr}(B) = \alpha \cdot 0 + 0 = 0.$$

Thus, $U$ is indeed a subspace. How do we find a basis for it? We'll start with the linearly-independent set $\emptyset$ in which we will add vectors in $U$ until we get a basis.

Notice that:

$$E_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \in U$$

and clearly $\{E_{1,2}\}$ is independent. Similarly:

$$E_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \in U$$

and we know that $\{E_{1,2}, E_{2,1}\}$ is linearly independent (since more generally $\{E_{i,j} : 1 \leq i, j \leq 2\}$ is the standard basis for $\mathcal{M}_2(\mathbb{F})$, so any subset of this must be independent).

Any $A \in \text{Span}(\{E_{1,2}, E_{2,1}\})$ is of the form:

$$A = \begin{bmatrix} 0 & b \\ c & 0 \end{bmatrix}$$

where $b, c \in \mathbb{F}$. But this doesn't capture all matrices of trace 0. For instance, we have:

$$E_{1,1} - E_{2,2} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \in U$$

but this cannot be expressed as a linear combination of $E_{1,2}$ and $E_{2,1}$. Thus, $\mathcal{B} = \{E_{1,2}, E_{2,1}, E_{1,1} - E_{2,2}\}$ is still a linearly-independent subset of $U$.

Are we finished? This is the point where knowing that $\dim(\mathcal{M}_2(\mathbb{F})) = 4$ really helps. We know that $\mathcal{B}$ is linearly independent. If it was not a spanning set for $U$, that would mean there is some other matrix $A \in U$ with $A \notin \text{Span}(\mathcal{B})$, and $\mathcal{B} \cup \{A\}$ would be linearly independent. This is a set with 4 elements, so by **Theorem 6.1 (iii)**, this forces it to be a basis for $\mathcal{M}_2(\mathbb{F})$. But we know $\mathcal{B} \cup \{A\} \subseteq U$, so this means in particular that $\mathcal{M}_2(\mathbb{F}) = \text{Span}(\mathcal{B}) \subseteq U$ and thus $U = \mathcal{M}_2(\mathbb{F})$, which we know is not true!

Therefore, we must have finished forming our basis for $U$ at the original three elements, which means $\mathcal{B} = \{E_{1,2}, E_{2,1}, E_{1,1} - E_{2,2}\}$ is indeed a basis for $U$. In particular, $\dim(U) = 3$.

**Exercise 6.9**

Find a basis for $\mathbb{R}^4$ which contains the vectors $\vec{v}_1 = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix}^T$ and $\vec{v}_2 = \begin{bmatrix} 0 & 2 & 0 & -4 \end{bmatrix}^T$.

**Exercise 6.10**

Find a basis for the subspace $U$ of $\mathcal{P}_3(\mathbb{F})$ spanned by:

$$\{1 + x, \quad x + x^2, \quad 2 + 3x + x^2, \quad x^2 + x^3, \quad 1 + 3x + 2x^2 + x^3\}.$$

**Exercise 6.11**

Find a basis for $\mathcal{P}_3(\mathbb{F})$ which consists only of vectors from the set $S = \{p(x) \in \mathcal{P}_3(\mathbb{F}) : p'(1) = 3\}$.

(Hint: here $S$ is infinite, so it would be best to find four independent vectors that are in $S$ rather than trying to remove redundant ones.)

**Exercise 6.12**

Consider the space $\mathcal{M}_3(\mathbb{F})$ of all $3 \times 3$ matrices over $\mathbb{F}$ and let:

$$U = \{A \in \mathcal{M}_3(\mathbb{F}) : \text{Tr}(A) = 0\}.$$

Find a basis for $U$ and compute $\dim(U)$.

## Lecture 7 Coordinates of a vector space

*This covers* ***Subsection 1.3.4*** *in the course notes.*

We saw that the notion of basis for a finite-dimensional vector space yields what we are (informally) calling a "coordinate system". In this lecture, we make this description more precise and complete. We saw that $\mathbb{F}^n$ has $n$ coordinates and verified that it indeed has dimension $n$ (as a vector space over $\mathbb{F}$). In this section, we will see that, once we choose a suitable basis and ordering, we can treat elements of an $n$-dimensional vector space $V$ exactly as elements of $\mathbb{F}^n$.

We begin with the result below which gives us further evidence that a basis for a finite-dimensional vector space $V$ over $\mathbb{F}$ is an appropriate notion for a coordinate system on $V$. So fix $V$ and let $\dim(V) = n$.

**Theorem 7.1 (Unique representation theorem)**

Let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be a basis for $V$. Then for every vector $\vec{v} \in V$, there exist *unique* scalars $\alpha_1, ..., \alpha_n \in \mathbb{F}$ for which:

$$\vec{v} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n.$$

We can think of this result as saying that a basis allows us to reach every $\vec{v} \in V$ in *one and only one* way using the vectors in the basis.

*Proof.* The fact that scalars $\alpha_1, ..., \alpha_n \in \mathbb{F}$ which satisfy the equality above exist simply follows from the fact that $\mathcal{B}$ is a spanning set. The important part of this theorem is the *uniqueness* part.

To prove this, suppose we had a second set of scalars $\beta_1, ..., \beta_n \in \mathbb{F}$ for which:

$$\vec{v} = \beta_1\vec{v}_1 + \cdots + \beta_n\vec{v}_n$$

Comparing this with the other expression, we obtain:

$$\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n = \beta_1\vec{v}_1 + \cdots + \beta_n\vec{v}_n$$

and moving the RHS to the LHS gives us:

$$(\alpha_1 - \beta_1)\vec{v}_1 + \cdots + (\alpha_n - \beta_n)\vec{v}_n = \vec{0}$$

Now since $\mathcal{B}$ is a linearly-independent set, it must follow that $\alpha_i - \beta_i = 0$ for every $1 \leq i \leq n$. In other words, $\alpha_i = \beta_i$ for every $i$. Hence, the scalars are in fact unique and this finishes the proof. □

This means if we have a known basis $\{\vec{v}_1, ..., \vec{v}_n\}$ for $V$, we can identify every element $\vec{v}$ of $V$ as an $n$-tuple $\begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^T \in \mathbb{F}^n$, with the correspondence given by:

$$\vec{v} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n \leftrightsquigarrow \begin{bmatrix}\alpha_1 & \cdots & \alpha_n\end{bmatrix}^T$$

There is however one subtle but important issue that needs to be addressed to have this correspondence make sense: an $n$-tuple is *ordered*, so we need to also assign an ordering on the basis elements so that we know where the coefficients get mapped to. This issue arises because sets don't specify any ordering of elements. For instance, $\mathcal{B} = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, ..., \vec{v}_n\}$ is the same as $\mathcal{B} = \{\vec{v}_2, \vec{v}_1, \vec{v}_3, ..., \vec{v}_n\}$.

**Definition 7.2 (Ordered basis)**

An **ordered basis** for $V$ is a basis $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ for $V$ along with a specification for how the vectors of $\mathcal{B}$ are being ordered.

We often will just write $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ with the implicit understanding that the vectors are in the order they are listed, but some texts will also use the notation $\mathcal{B} = (\vec{v}_1, ..., \vec{v}_n)$.

So for example, $(\vec{v}_1, \vec{v}_2, \vec{v}_3, ..., \vec{v}_n)$ is an ordered basis, $(\vec{v}_2, \vec{v}_1, \vec{v}_3, ..., \vec{v}_n)$ is a *different* ordered basis, but they represent the same unordered basis for $V$.

**Remark 7.3 (Standard ordered bases)**

We have defined and worked with *standard* bases for $\mathbb{F}^n$, $\mathcal{P}_n(\mathbb{F})$, and $\mathcal{M}_{m\times n}(\mathbb{F})$. We will equip ordering as follows:

- The standard basis $\{\vec{e}_1, ..., \vec{e}_n\}$ for $\mathbb{F}^n$ will be ordered as written.
- Similarly the standard basis $\{1, x, x^2, ..., x^n\}$ for $\mathcal{P}_n(\mathbb{F})$ will be ordered as written.
- For the standard basis $\{E_{i,j} : 1 \le i \le m, 1 \le j \le n\}$ for $\mathcal{M}_{m\times n}(\mathbb{F})$, we will order it in ascending "lexicographical" order of the pair $(i, j)$. That is:

$$E_{1,1}, E_{1,2}, ..., E_{1,n}, \quad E_{2,1}, E_{2,2}, ..., E_{2,n}, \quad ..., \quad E_{m,1}, E_{m,2}, ..., E_{m,n}.$$

We will call these the **standard ordered bases** for the respective spaces.

We can now make our correspondence more precise:

**Definition 7.4 (Coordinate vector)**

Suppose $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ is an ordered basis for $V$. For every $\vec{v} \in V$, we define the **coordinate vector** of $\vec{v}$ with respect to $\mathcal{B}$ as:

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \in \mathbb{F}^n$$

where $\alpha_1, ..., \alpha_n$ are the unique coefficients satisfying $\vec{v} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n$.

Notice here how the order on the vectors of $\mathcal{B}$ is important. If for instance $n = 3$ and we set $\mathcal{C} = \{\vec{v}_1, \vec{v}_3, \vec{v}_2\}$, then we would have:

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}, \quad \text{and} \quad [\vec{v}]_{\mathcal{C}} = \begin{bmatrix} \alpha_1 \\ \alpha_3 \\ \alpha_2 \end{bmatrix}$$

which in general will be different vectors.

Coordinate vectors give us a nice way to study and work with our vector space in terms of coordinates as we do with $\mathbb{F}^n$. To make use of the full extent of $\mathbb{F}^n$, we should ensure that addition and scalar multiplication in $V$ corresponds to addition and scalar multiplication on $\mathbb{F}^n$ via the representation:

**Theorem 7.5 (Coordinate vectors preserve + and ·)**

Choose an ordered basis $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ of $V$. Then for all $\vec{v}, \vec{w} \in V$ and $\alpha \in \mathbb{F}$, we have:

$$[\vec{v} + \vec{w}]_{\mathcal{B}} = [\vec{v}]_{\mathcal{B}} + [\vec{w}]_{\mathcal{B}}$$
$$[\alpha\vec{v}]_{\mathcal{B}} = \alpha[\vec{v}]_{\mathcal{B}}$$

Moreover, we have for each $1 \leq i \leq n$ that $[\vec{v}_i]_{\mathcal{B}} = \vec{e}_i \in \mathbb{F}^n$, where $\vec{e}_i$ is the $i^{\text{th}}$ standard basis vector for $\mathbb{F}^n$.

*Proof.* Exercise. □

The main takeaway from this is the following: you can view vectors in a finite-dimensional vector space as vectors in $\mathbb{F}^n$ by choosing a suitable coordinate system (i.e. an ordered basis) for $V$.

**Example 7.6**

Consider $\mathcal{P}_4(\mathbb{F})$ and the standard ordered basis $\mathcal{E} = \{1, x, x^2, x^3, x^4\}$. Let us put the elements of $\mathcal{E}$ in the order they are listed to make it an ordered basis. Then we will have:

$$[a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4]_{\mathcal{E}} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} \in \mathbb{F}^5$$

In certain situations, we will be given a more "complicated" coordinate system and must find the right coefficients to express a certain vector in terms of that system. Often this just boils down to solving a system of linear equations. Let's see an example of what we mean.

**Example 7.7**

Consider the vector space $\mathcal{P}_2(\mathbb{R})$ set:

$$\mathcal{B} = \left\{1 - x, 1 + x + x^2, 1 - x + x^2\right\}.$$

We will accept without proof that $\mathcal{B}$ is indeed an ordered basis for $\mathcal{P}_2(\mathbb{R})$ (it may be worth verifying as an exercise).

Now consider:

$$p(x) = 3 + 3x + 5x^3.$$

We wish to determine how we can write $\vec{x}$ in terms of the ordered basis $\mathcal{B}$. In other words, we want to find:

$$[p(x)]_{\mathcal{B}} = \left[3 + 3x + 5x^2\right]_{\mathcal{B}} \in \mathbb{R}^3.$$

This means we need to find out which $a, b, c \in \mathbb{R}$ satisfy the equality:

$$a(1 - x) + b(1 + x + x^2) + c(1 - x + x^2) = 3 + 3x + 5x^2$$

$$\implies (a + b + c) + (-a + b - c)x + (b + c)x^2 = 3 + 3x + 5x^2.$$

Comparing the coefficients of the polynomials, we obtain a system of three linear equations in three unknowns:

$$\begin{cases} a + b + c & = 3 \\ -a + b - c & = 3 \\ b + c & = 5 \end{cases}$$

From MATH 136, we know that we can solve the system of equations by using row-reduction. Form the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ -1 & 1 & -1 & 3 \\ 0 & 1 & 1 & 5 \end{array}\right]$$

and applying row reduction operations (which we will leave as an exercise), we will obtain:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

Hence, we have $a = -2$, $b = 3$, and $c = 2$. In other words:

$$[p(x)]_{\mathcal{B}} = \left[3 + 3x + 5x^2\right]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \\ 2 \end{bmatrix}$$

**Exercise 7.8**

Consider the vector space $\mathcal{P}_2(\mathbb{R})$. Let $p(x) = (1-x)^2$, $q(x) = 2x(1-x)$, and $r(x) = x^2$. Show that $\mathcal{B} = \{p(x), q(x), r(x)\}$ is an ordered basis for $\mathcal{P}_2(\mathbb{R})$ and find an expression for the vector $[ax^2 + bx + c]_{\mathcal{B}} \in \mathbb{R}^3$ in terms of $a$, $b$, and $c$.

We come to a useful fact about spanning and independence for coordinate vectors (we shall omit the proof for the sake of time).

**Proposition 7.9 (Coordinate vectors preserve spanning + indep.)**

Choose an ordered basis $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ for $V$.

Let $S = \{\vec{w}_1, ..., \vec{w}_k\}$ be any set of vectors in $V$ and let $S_{\mathcal{B}} = \{[\vec{w}_1]_{\mathcal{B}}, ..., [\vec{w}_k]_{\mathcal{B}}\} \subseteq \mathbb{F}^n$. Then:

**(i)** $S$ is a spanning set for $V$ if and only if $S_{\mathcal{B}}$ is a spanning set for $\mathbb{F}^n$.

**(ii)** $S$ is a linearly-independent subset of $V$ if and only if $S_{\mathcal{B}}$ is a linearly-independent subset of $\mathbb{F}^n$.

Using the theorem above, we obtain another advantage of working with coordinate vectors. Namely, we shall have an easier time determining if certain subsets are spanning and/or linearly independent. Let us see an example of this.

**Example 7.10**

Consider the vector space $\mathcal{P}_2(\mathbb{F})$ and let $S = \{1 - x - x^2, 2 + 3x + 4x^2, 5 - 2x + x^2\}$. We wish to determine if $S$ is linearly independent. To do so, let $\mathcal{E} = \{1, x, x^2\}$ be the standard ordered basis. Then:

$$\vec{x}_1 = [1 - x - x^2]_{\mathcal{E}} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \quad \vec{x}_2 = [2 + 3x + 4x^2]_{\mathcal{E}} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\vec{x}_3 = [5 - 2x + x^2]_{\mathcal{E}} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

and $S$ is linearly independent if and only if $\{\vec{x}_1, \vec{x}_2, \vec{x}_3\}$ is linearly independent in $\mathbb{F}^3$. We know from 136 how to determine if this is so: place $\vec{x}_1, \vec{x}_2, \vec{x}_3$ as columns in a matrix like so:

$$A = \begin{bmatrix} 1 & 2 & 5 \\ -1 & 3 & -2 \\ -1 & 4 & 1 \end{bmatrix}.$$

Upon performing row reduction to place $A$ in RREF (left as an exercise), we find:

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

So the rank of $A$ is 3, therefore the columns must be linearly independent, and hence so must $S$.

Thus, we have essentially turned a MATH 235 problem (of a given abstract vector space) into a MATH 136 problem (of $\mathbb{F}^n$). Let us see another example.

**Example 7.11**

Consider the vector space $\mathcal{M}_2(\mathbb{F})$ and the set:

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \right\}.$$

We must determine if $S$ is spanning. For this, we can let $\mathcal{E} = \{E_{1,1}, E_{1,2}, E_{2,1}, E_{2,2}\}$ be the standard ordered basis for $\mathcal{M}_2(\mathbb{F})$ and obtain:

$$\vec{x}_1 = \left[ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{x}_2 = \left[ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \right]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{x}_3 = \left[ \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \right]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\vec{x}_4 = \left[ \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \right]_{\mathcal{E}} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 3 \end{bmatrix}, \quad \vec{x}_5 = \left[ \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \right]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix}.$$

Then $S$ is a spanning set for $\mathcal{M}_2(\mathbb{F})$ if and only if $\{\vec{x}_1, ..., \vec{x}_5\}$ is a spanning set for $\mathbb{F}^4$. This is true precisely when the matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 3 & 2 \end{bmatrix}$$

has $\text{Col}(A) = \mathbb{F}^4$. When we compute the RREF of $A$, we obtain:

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Notice there are only three pivots and they appear on columns 1, 2, and 3. Therefore, $\text{Col}(A)$ has basis:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

(the first three columns of $A$, and the other two columns are redundant). Since $\dim(\mathbb{F}^4) = 4$, $\text{Col}(A) \neq \mathbb{F}^4$ and therefore $S$ is *not* a spanning set for $\mathcal{M}_2(\mathbb{F})$.

**Exercise 7.12**

Consider $\mathcal{M}_2(\mathbb{F})$ and the set:

$$S = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \right\}$$

(this was seen in **Example 6.5** of the last lecture). Prove that the set $S$ from the example is indeed a spanning set for $\mathcal{M}_2(\mathbb{F})$ using what we have learned in this lecture.

**Exercise 7.13**

Determine whether the following set of polynomials is linearly independent in $\mathcal{P}_2(\mathbb{F})$:

$$S = \{1 + 4x + 7x^2, \quad 2 + 5x + 8x^2, \quad 3 + 6x + 9x^2\}$$

## Lecture 8 Linear maps between vector spaces

*This covers **Section 2.1** in the course notes.*

Now that we have studied properties of vector spaces themselves, we shall next look at how two different spaces can "interact". To make this precise, we want to take two vector spaces $V$ and $W$ over (a common field) $\mathbb{F}$ and see how we can associate to each vector $\vec{v} \in V$ a corresponding vector $\vec{w} \in W$.

We know a way in which this correspondence is achieved: by finding a function $f : V \to W$. The job of $f$ is to then associate to each $\vec{v} \in V$ a vector $f(\vec{v}) \in W$. But a function is nothing but a map between sets. We want $f$ to also preserve the vector space structures belonging to $V$ and $W$. In order to achieve this, we define:

**Definition 8.1 (Linear map and linear operator)**

Let $V$ and $W$ be two vector spaces over $\mathbb{F}$. A **linear map** (or **linear transformation**) from $V$ to $W$ is a function $T : V \to W$ which satisfies the following properties:

**(i)** For all $\vec{v}_1, \vec{v}_2 \in V$, $T(\vec{v}_1 + \vec{v}_2) = T(\vec{v}_1) + T(\vec{v}_2)$.

**(ii)** For all $\vec{v} \in V$ and $\alpha \in \mathbb{F}$, $T(\alpha\vec{v}) = \alpha T(\vec{v})$.

We will often drop the brackets in $T(\vec{v})$ and just write $T\vec{v}$ if it is clear that $T$ is being applied to only one vector. This will make things more readable later on.

A linear map $T : V \to V$ (both the domain and codomain are the same vector space $V$) is often called a **linear operator** on $V$.

As we shall see, linear maps can tell us a lot about the vector spaces and subspaces we have encountered already. Particularly, if we know about a vector space $V$ but don't know a lot about another vector space $W$, we can often learn more about $W$ by finding a linear map $T$ going from $V$ to $W$ (or even vice versa).

We will also see later on that linear maps play an even greater role. For example, we can also study a given vector space $V$ by looking at the linear maps from $V$ to itself. First, we have the following useful observation which can help us make our proofs of linearity more concise:

**Proposition 8.2**

Let $T : V \to W$ be a map between two vector spaces $V$ and $W$ over $\mathbb{F}$. Then $T$ is linear if and only if for all $\vec{v}_1, \vec{v}_2 \in V$ and all $\alpha \in \mathbb{F}$:

$$T(\alpha\vec{v}_1 + \vec{v}_2) = \alpha T\vec{v}_1 + T\vec{v}_2$$

In other words, we can combine conditions **(i)** and **(ii)** in the linearity definition into one line (the same way we made the subspace test more concise earlier).

*Proof.* Exercise. □

We will now look at a gallery of examples.

**Example 8.3**

First, some examples which work in general vectors spaces. Let $V$ and $W$ be any two vector spaces over $\mathbb{F}$.

**(1)** We have the **identity map** $\mathrm{id}_V : V \to V$ which is given by $\mathrm{id}_V(\vec{v}) = \vec{v}$. We can also abbreviate it to id if the vector space $V$ is understood. It is also common to denote it as $I$ or $I_V$. Obviously this is linear.

**(2)** We also have the **zero map** $O : V \to W$ which is given by $O(\vec{v}) = \vec{0}_W$ for all $\vec{v} \in V$, where $\vec{0}_W$ is the zero vector in $W$. It is clear that this is also linear.

**(3)** Suppose $U$ is a subspace of $V$. Then $U$ is itself a vector space of course, so it makes sense to define the map $\iota : U \to V$ given by $\iota(\vec{u}) = \vec{u}$ for all $\vec{u} \in U$. We call $\iota$ the **inclusion map** of $U$ into $V$.

There is essentially no difference between the inclusion map of $U$ into $V$ and the identity map on $U$, apart from the specification of the codomain. As we shall see later though, this specification can be important.

**Example 8.4**

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be given by

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

That is, $T$ takes a point in $\mathbb{R}^3$ and projects it onto the $xy$-plane sitting inside $\mathbb{R}^3$ (where the $xy$-plane is identified as a copy of $\mathbb{R}^2$). Notice that for any $x_1, y_1, z_1, x_2, y_2, z_2 \in \mathbb{R}$ and $\alpha \in \mathbb{R}$:

$$\begin{aligned} T\left(\alpha\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}\right) &= T\begin{bmatrix} \alpha x_1 + x_2 \\ \alpha y_1 + y_2 \\ \alpha z_1 + z_2 \end{bmatrix} \\ &= \begin{bmatrix} \alpha x_1 + x_2 \\ \alpha y_1 + y_2 \end{bmatrix} \\ &= \alpha\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \\ &= \alpha T\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + T\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \end{aligned}$$

Hence, $T$ is linear.

**Example 8.5**

Let $D : \mathcal{P}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})$ be given by $D(p(x)) = p'(x)$. More precisely:

$$D\left(\sum_{k=0}^{n} a_k x^k\right) = \sum_{k=0}^{n-1} k a_k x^k.$$

We call this the **differentiation map**. The fact that it's linear follows from the linearity property of differentiation.

Notice also that we can restrict the domain and codomain: since $p(x) \in \mathcal{P}_n(\mathbb{F})$ implies $D(p(x)) \in \mathcal{P}_{n-1}(\mathbb{F})$, we can restrict $D$ to $\mathcal{P}_n(\mathbb{F})$ to give a linear operator $D : \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$ on $\mathcal{P}_n(\mathbb{F})$.

**Example 8.6**

Let $\mathrm{Tr} : \mathcal{M}_n(\mathbb{F}) \to \mathbb{F}$ be the trace map:

$$\mathrm{Tr}([a_{i,j}]) = \sum_{k=1}^{n} a_{k,k}.$$

We have then that for any $A = [a_{i,j}], B = [b_{i,j}] \in \mathcal{M}_n(\mathbb{F})$ and $\alpha \in \mathbb{F}$:

$$\begin{aligned}
\mathrm{Tr}(\alpha A + B) &= \mathrm{Tr}([\alpha a_{i,j} + b_{i,j}]) \\
&= \sum_{k=1}^{n} (\alpha a_{k,k} + b_{k,k}) \\
&= \alpha \sum_{k=1}^{n} a_{k,k} + \sum_{k=1}^{n} b_{k,k} \\
&= \alpha\,\mathrm{Tr}(A) + \mathrm{Tr}(B)
\end{aligned}$$

So Tr is linear.

**Exercise 8.7**

Show that all of the following maps are linear:

**(a)** For $\theta \in [0, 2\pi)$, the **rotation map** $T_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ given by:

$$T_\theta \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}.$$

**(b)** The **integration map** $V : \mathcal{P}(\mathbb{F}) \to \mathcal{P}(\mathbb{F})$ given by:

$$V(p(x)) = \int_0^x p(t)\,dt.$$

**(c)** The **transpose map** $\tau : \mathcal{M}_{m\times n}(\mathbb{F}) \to \mathcal{M}_{n\times m}(\mathbb{F})$ given by:

$$\tau(A) = A^T$$

where $A^T$ is the transpose of $A$.

**Exercise 8.8**

Suppose $V$ is a vector space over $\mathbb{F}$ and $U$ is a subspace. Let $T : V \to U$ be given by:

$$T\vec{v} = \begin{cases} \vec{v}, & \vec{v} \in U \\ 0, & \vec{v} \notin U \end{cases}$$

for all $\vec{v} \in V$. Is $T$ a linear map? Either prove it or give a counterexample.

**Exercise 8.9**

Let $\det : \mathcal{M}_n(\mathbb{F}) \to \mathbb{F}$ be the determinant map. Is det a linear map for all $n \geq 1$? Either prove it or give a counterexample.

(Hint: if you think it is *not* a linear map for all $n \geq 1$, a nice place to find a counterexample would be $\mathcal{M}_2(\mathbb{F})$.)

**Exercise 8.10**

Consider the space $\mathcal{C}([0,1])$ of continuous functions from $[0,1]$ to $\mathbb{R}$. Recall that in a past exercise, we defined:

$$\mathcal{C}^1([0,1]) = \{f \in \mathcal{C}([0,1]) : f \text{ is differentiable and } f' \text{ is continuous}\}$$

and saw that this is a subspace of $\mathcal{C}([0,1])$.

**(a)** Show that the map $D : \mathcal{C}^1([0,1]) \to \mathcal{C}([0,1])$ given by $D(f) = f'$ is linear.

**(b)** Find an example of some $f \in \mathcal{C}^1([0,1])$ where $D(f) = f' \notin \mathcal{C}^1([0,1])$.

Let us now prove two important facts about linear maps. The first one we give is sometimes used as a sanity check to see if a given map is *not* linear.

**Proposition 8.11 (Linear maps send zero to zero)**

If $T : V \to W$ is a linear map between vector spaces $V$ and $W$ over $\mathbb{F}$, then $T\vec{0}_V = \vec{0}_W$ where $\vec{0}_V$ and $\vec{0}_W$ are the zero vectors of $V$ and $W$, respectively.

*Proof.* $T\vec{0}_V = T(0 \cdot \vec{0}_V) = 0 \cdot T\vec{0}_V = \vec{0}_W$. □

Of course, we can't conclude from $T\vec{0}_V = \vec{0}_W$ alone that $T$ is linear. For instance, $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = x^2$ sends 0 to 0, but is not linear since $T(2) = T(2 \cdot 1) = 4$ while $2T(1) = 2 \cdot 1 = 2$.

The next proposition gives us a powerful way to define and characterize linear maps on finite-dimensional spaces. In particular, it makes checking if two linear maps are equal much easier in a lot of situations.

**Proposition 8.12 (Linear maps are determined on a basis)**

Let $V$ and $W$ be vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be any basis for $V$.

Then for any $\vec{w}_1, ..., \vec{w}_n \in W$, there exists a *unique* linear map $T : V \to W$ satisfying $T\vec{v}_i = \vec{w}_i$ for all $1 \leq i \leq n$.

*Proof.* To see that $T$ exists, consider the mapping $T : V \to W$ given as follows: for every $\vec{u} \in V$ with $\vec{u} = \alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n$, we recall that the $\alpha_i$'s are uniquely determined by what $\vec{u}$ is chosen. So set $T(\vec{u}) = \alpha_1\vec{w}_1 + \cdots + \alpha_n\vec{w}_n$ and it is clear that this is a well-defined map.

To show this is linear, let $\vec{u}, \vec{v} \in V$ with:

$$\vec{u} = \alpha_1\vec{v}_1 + \cdots + \alpha_1\vec{v}_n$$
$$\vec{v} = \beta_1\vec{v}_1 + \cdots + \beta_n\vec{v}_n$$

and let $\gamma \in \mathbb{F}$. Then:

$$\gamma\vec{u} + \vec{v} = (\gamma\alpha_1 + \beta_1)\vec{v}_1 + \cdots + (\gamma\alpha_n + \beta_n)\vec{v}_n$$

and each coefficient above is also uniquely determined of course, and we have:

$$\begin{aligned} T(\gamma\vec{u} + \vec{v}) &= (\gamma\alpha_1 + \beta_1)\vec{w}_1 + \cdots + (\gamma\alpha_n + \beta_n)\vec{w}_n \\ &= \gamma(\alpha_1\vec{w}_1 + \cdots + \alpha_n\vec{w}_n) + (\beta_1\vec{w}_1 + \cdots + \beta_n\vec{w}_n) \\ &= \gamma T\vec{u} + T\vec{v} \end{aligned}$$

Thus, $T$ is linear.

The uniqueness part is left as an exercise. □

**Corollary 8.13 (Linear maps are equal iff they agree on a basis)**

Let $V$ and $W$ be vector spaces over $\mathbb{F}$ with $\dim(V) = n$ and let $\mathcal{B} = \{\vec{v}_1, ..., \vec{v}_n\}$ be any basis for $V$.

If $S : V \to W$ and $T : V \to W$ are any two linear maps, then $S = T$ if and only if $S\vec{v}_i = T\vec{v}_i$ for every $1 \leq i \leq n$.

*Proof.* Exercise. □

Finally, let us collect some algebraic properties about linear maps in general, which will prove useful in a later section.

**Proposition 8.14 (Sum and composition of linear maps)**

Let $V$ and $W$ be vector spaces over $\mathbb{F}$. Let $S, T : V \to W$ be two linear maps from $V$ to $W$ and let $\alpha \in \mathbb{F}$. Then the maps $S+T : V \to W$ and $\alpha S : V \to W$, which are given by:

$$(S+T)\vec{v} = S\vec{v} + T\vec{v} \ \ \forall \vec{v} \in V$$

$$(\alpha S)\vec{v} = \alpha S\vec{v} \ \ \forall \vec{v} \in V$$

are also linear maps. In particular, if $\mathcal{L}(V, W)$ denotes the set of all linear maps from $V$ to $W$, then $\mathcal{L}(V, W)$ is a vector space under the above addition and scalar multiplication operations.

Moreover, if $U$ is another vector space over $\mathbb{F}$ and $R : U \to V$ is a linear map, then the composition function $S \circ R : U \to W$ given by:

$$(S \circ R)\vec{u} = S(R\vec{u}) \ \forall \vec{u} \in U$$

is a linear map as well.

*Proof.* Exercise. □

**Remark 8.15**

In the above, we often write $S \circ R$ as just $SR$ for brevity, and write $S(R\vec{u}) = SR\vec{u}$. Also, composition of linear maps is often just called **multiplication**. The reason for this will become apparent soon.

**Practice Problems**

Here are some practice problems covering what we have done this week.

**P1** Consider the space $\mathcal{M}_2(\mathbb{F})$ of $2 \times 2$ matrices over $\mathbb{F}$. We say $E \in \mathcal{M}_2(\mathbb{F})$ is **idempotent** when $E^2 = E$.

Find a basis for $\mathcal{M}_2(\mathbb{F})$ consisting of idempotents. That is, find a basis $\mathcal{B} = \{E_1, E_2, E_3, E_4\}$ for $\mathcal{M}_2(\mathbb{F})$ such that $E_i^2 = E_i$ for every $1 \leq i \leq 4$.

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**P2** Consider the space $\mathcal{P}_2(\mathbb{R})$ and let:

$$U = \{p(x) \in \mathcal{P}_2(\mathbb{R}) : p(1-x) = p(1+x)\}.$$

**(a)** Verify that $U$ is a subspace of $\mathcal{P}_2(\mathbb{R})$.

**(b)** Find a basis for $U$.

**(c)** Use part **(b)** to compute $\dim(U)$.

---

**P3** Consider the vector space $\mathcal{P}_3(\mathbb{F})$ having standard ordered basis $\{1, x, x^2, x^3\}$. Let $p(x) \in \mathcal{P}_3(\mathbb{F})$ have the form:

$$p(x) = a + bx + cx^2 + dx^3$$

for some coefficients $a, b, c, d \in \mathbb{F}$. For each ordered basis $\mathcal{B}$ of $\mathcal{P}_3(\mathbb{F})$ listed below, find $[p(x)]_{\mathcal{B}}$ (you can assume without proof that they are indeed bases):

**(a)** $\mathcal{B} = \{1, 1+x, x+x^2, x^2+x^3\}$.

**(b)** $\mathcal{B} = \{1, x, x^2-1, x^3-3x\}$.

**(c)** $\mathcal{B} = \{(1-x)^3, 3x(1-x)^2, 3x^2(1-x), x^3\}$.

---

**P4** Let $x \in \mathbb{R}$ be given and let:

$$S = \left\{ \begin{bmatrix} x \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ x \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} \right\} \subseteq \mathbb{R}^3.$$

**(a)** For which values of $x$ is $S$ a linearly-*dependent* subset of $\mathbb{R}^3$?

**(b)** For each such value of $x$, compute the dimension of the corresponding subspace spanned by $S$.

---

**P5** Determine if the following set is a spanning set for $\mathcal{P}_3(\mathbb{F})$:

$$S = \left\{x^3 + x^2 + x + 1, \quad x^3 + 2x^2 + x, \quad 2x^3 + 3x^2 + 2x + 1, \quad x^3 + x^2 + 2x + 1\right\}.$$

---

**P6** Let $V$ and $W$ be vector spaces (not necessarily finite- dimensional) over $\mathbb{F}$ and let $T : V \to W$ be a linear map. Let $\vec{v}_1, ..., \vec{v}_n \in V$ be distinct vectors and set $\vec{w}_i = T\vec{v}_i$ for each $1 \leq i \leq n$.

**(a)** Show that if $\vec{w}_1, ..., \vec{w}_n$ are distinct and $\{\vec{w}_1, ..., \vec{w}_n\}$ is linearly independent, then the set $\{\vec{v}_1, ..., \vec{v}_n\}$ is also linearly independent.

**(b)** If $\{\vec{v}_1, ..., \vec{v}_n\}$ is linearly independent, is the same necessarily true for $\{\vec{w}_1, ..., \vec{w}_n\}$? Prove or disprove.